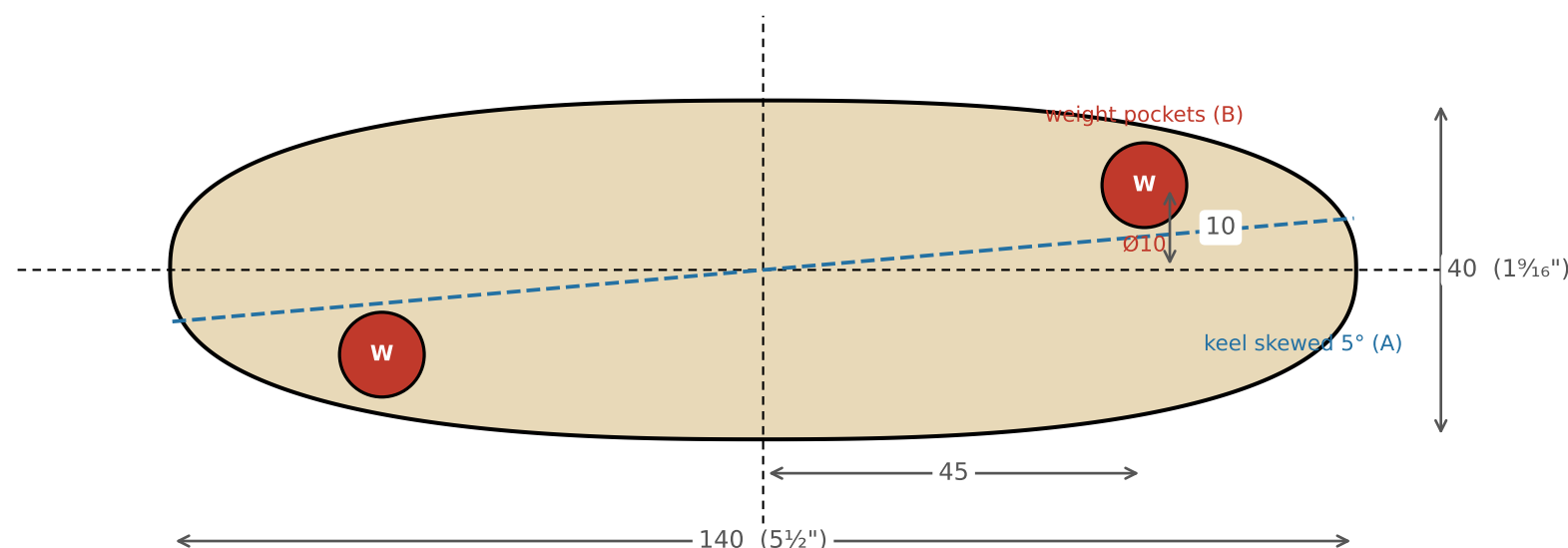


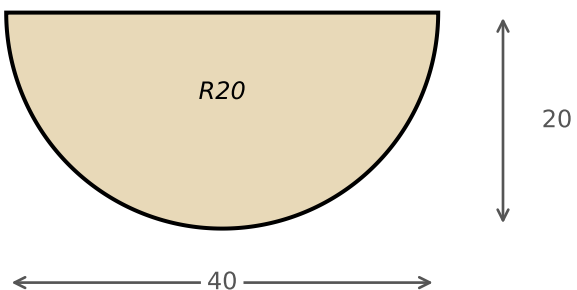
SHORT 8 — RATTLEBACK — BUILD PLAN

All dimensions in mm. Based on the proven 140 × 40 mm wooden rattleback.

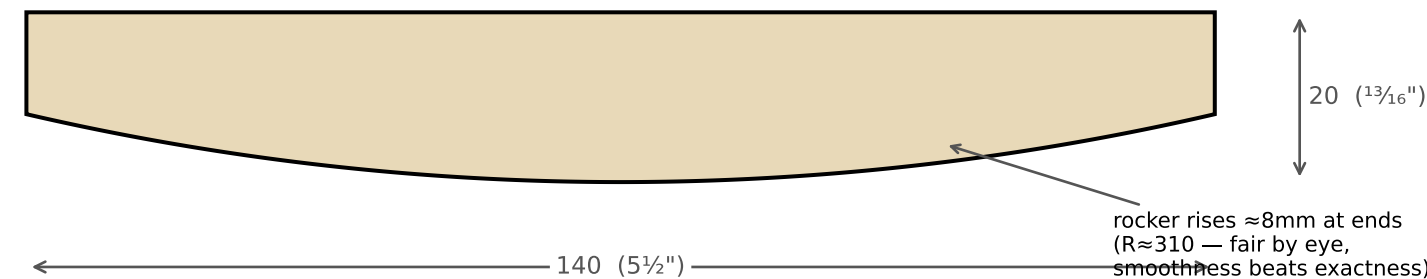
TOP VIEW



END SECTION (at middle)



SIDE VIEW



BUILD SEQUENCE

1. Blank: 140 × 40 × 20 mm hardwood (maple, beech or walnut).
2. Cut the canoe outline (top view).
3. Round the bottom across the width → semicircular section (end view).
4. Carve the lengthwise rocker — ends rise ≈8 mm. Fair it dead smooth; no flats or facets.
5. Add the skew — pick ONE:
 - A: drift the keel line 6 mm off center at each end, opposite sides (5°).
 - B: drill Ø10 × 10 deep pockets at W, pack with lead shot / BBs, epoxy, plug flush.
6. Sand to 220, oil or wax. A slick bottom helps.

TEST (glass or polished level bench)

- Sit it down: it must rest level.
- Spin both ways: one way rattles → stops → reverses. The other way just spins.
- No reversal? Add more skew / shift the weights and retest. Tuning is part of the build — film it.